Qualitative approaches to digital trace data

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Abstract
In this paper, we argue that research strategies captured under the umbrella term qualitative approaches are essential for the successful social science use of large-scale data platforms or infrastructures such as social media, search engines and sensors like mobile phones. A key point here is that social science needs to devise methods that properly account for the social situation responsible for generating the data, what we call the data situation. Flexibility in gathering qualitative research data, exploratory and interpretative analysis, and attentiveness to context are essential elements for understanding the data situation across social media data, search data, and sensory data. We present four research tasks where qualitative approaches, in our view, are essential for the exploitation of large-scale data infrastructures for social science research: *casing* of the phenomenon under study, *grounding* the interpretation of the social process, *theorizing* the phenomenon in light of surprising results and *validating* both the measures and analysis. As we argue, without proper casing we risk mis-specifying our object of study; without grounding we risk misunderstanding our observations; without active theorizing we can't exploit surprising findings; and without qualitative validation we can't trust quantitative measures that build upon qualitative data. We substantiate this need for qualitative approaches in digital social research via examples from our own projects using social media data and sensor devices.

1. Introduction

The activity traces recorded by digital infrastructures —so-called digital trace data—typically capture social life at particular moments and within specific situations. When we post on social media, we do so in response to specific events, in the context of other messages, and with a particular audience in mind. When we search online, we are usually pursuing a concrete goal—whether it is finding entertainment or checking the opening hours of the local supermarket. Even the traces from our mobile phones record our movement through physical space, tracking our commute from home to a specific workplace and noting stops for coffee along the way. From this perspective, the digital data revolution is a revolution in data about social situations. Digital trace data share many characteristics with qualitative data: they capture people's actions and utterances in specific contexts—and this situatedness is what gives those actions their meaning and effect. Additionally, digital trace data are often composed of more or less unstructured bits of text, which makes them open-ended in terms of their meaning and interpretability, calling for interpretative analysis. Yet, the scale of digital trace data is typically much larger than that of traditional qualitative data.

Although digital in form and large in scale, trace data are qualitative because they consist of unstructured text produced in concrete social situations, where meaning is context-dependent and cannot be reduced to predefined variables. Despite this, a dominant perspective in the social sciences holds that the digital data revolution is primarily relevant to quantitative research. In his influential book *Bit by Bit*, Salganik argues that

this revolution mainly concerns the quantitative social sciences (Mayer-Schönberger and Cukier 2013; Salganik 2017). This view is a common assumption in both research and education (for a few exceptions, see Kozinets 2015; Marres 2017; Rogers 2013). While the digital revolution has certainly been regarded as an important *object* of study for qualitative researchers (Beaulieu 2004; Miller 2011), it has not been equally embraced as a *source* of data or as a site for methodological innovation in qualitative research. Courses in digital social science typically emphasize students' ability to code, apply machine learning techniques, and analyze large datasets. However, they seldom engage with foundational qualitative concerns such as case construction, interpretation, and theorization. This gap creates a critical problem: digital trace data are often reappropriated as purely quantitative data without acknowledging their fundamentally qualitative character. This leads to a range of issues, including the misidentification of the object of study, measurement errors, and the application of ill-fitting theoretical frameworks.

In this paper, we argue that research strategies captured under the umbrella term qualitative approaches are essential for the successful social scientific *use* of digital trace data. We then outline four distinct qualitative tasks that, in our view, are necessary for social scientific research using large-scale data platforms or infrastructures such as social media, search engines, and sensors like mobile phones: *casing* the phenomenon under study, *grounding* interpretations of social processes, *theorizing* the phenomenon in light of surprising results, and *validating* both quantitative measures and analyses. As we argue, without proper casing we risk mis-specifying our object of study; without grounding we risk misunderstanding our observations; without active theorizing we can't exploit surprising findings; and without qualitative validation we can't trust our quantitative measures that build upon qualitative data. We substantiate this need for qualitative approaches in social science research with digital trace data via examples from our own projects using social media data and sensor devices.

2. Digital trace data is qualitative data

In this section we argue and demonstrate why digital trace data despite its scale and digital nature is in many regards qualitative data. We introduce the concept of the *data situation* to foreground the social and technical conditions under which digital trace data is generated and made available for analysis (Marres 2017; Ruppert, Law, and Savage 2013). Analyzing the data situations responsible for generating one's data helps clarify both how to understand the digital data and make informed decisions around that qualitative data collection efforts and analysis necessary for producing sound social science research. We will use the concept of the data situation to describe three prominent data sources: social media data, search data and sensory data.

On the more technical side of the data situation are considerations regarding the extent to which data generation is influenced by algorithms, user interfaces, and other design features (Lazer et al. 2021; Salganik 2017). However, the technical aspects we focus on here relate more to the character of the data available to the researcher. Specifically, we consider two key factors: the *open or closed* nature of the data, and the *degree of datafication* of the social situation. Both of these factors influence the types of qualitative data available and the methods that can be applied. By *open* or *closed*, we refer to something akin to the distinction between an open-ended survey question—where respondents answer in their own words—and a closed-ended item, where responses are limited to predefined options. This distinction also applies to digital platforms: for example, a Facebook "like" represents a closed format, while a comment constitutes an open format. Open formats allow for—and partly demand—more qualitative interpretation than closed data formats do.

The *degree of datafication* concerns the extent to which the information required to understand a situation is available as digital data (for more on datafication of social processes see Lupton 2016; Mayer-Schönberger and Cukier 2013; Van Dijck 2014). In other words, is it possible to grasp the situation through qualitative analysis of digital trace data alone, or is it necessary to collect traditional qualitative data such as interviews or situational observations? A phone trace that simply records a user's latitude and longitude has a low degree of datafication. The person might be engaged in an important social ritual that strengthens ties with peers from their study program, but all we know from the data is that they are located somewhere in the northern part of Greater Copenhagen. By contrast, access to all the comments and posts in a refugee support group in Denmark reflects a relatively high degree of datafication. In such cases, we can often assume that we can reconstruct from our data a similar relational context to that of the users interacting in the forum. Unlike phone sensor data, the digital traces on social media are often an integral part of the social situation itself.

Search data: from Google and other search services has been described as "the most important data set ever collected on the human psyche" (Stephens-Davidowitz 2017). While this may be a bold claim, the data is undeniably rich and powerful. A brief reflection on one's own search history makes it clear why: searches reveal worries, practical concerns, and desires—often including things we might never disclose to others. In this sense, search data constitutes unobtrusive, large-scale recordings of human inner life. It offers a unique resource for understanding both ourselves and one another. Yet, to use search data responsibly and effectively, we must pay careful attention to the specific conditions and contexts under which it is generated.

Firstly, as Salganik and many others have noted, search data is non-representative, incomplete, and subject to algorithmic confounders (Salganik 2017). Not everyone searches for the same topics, which makes it difficult to generalize findings to the broader population. Moreover, we typically have little to no demographic or contextual information about the individuals conducting the searches, which limits the kinds of analyses we would ideally like to perform. A further source of incompleteness stems from the nature of the data made available to researchers. Instead of access to full search strings or individual-level search records, we are often limited to aggregate statistics—typically frequencies of single words, distributed over time and across regions. While one might argue that search data resembles open-ended survey items in that people can use their own words, the way this information is shared with researchers is usually highly constrained. This thinness of the data has important consequences for how we can interpret it and what additional data we need to support interpretations[1].

The low *degree of datafication* of the social situation results in fundamental uncertainties about what search data actually represents. Precisely because searches serve a wide range of practical purposes and our data trace is limited, it is difficult to assume that a query for *X* straightforwardly means *Y*. To take a simple example: if I search for "Facebook," is it because I am concerned about the platform, because I want information about the platform, or simply because I am lazy and prefer to type "Facebook" into Google rather than enter the URL directly? These are analytically distinct behaviors, but they are indistinguishable in the data. Therefore, we must first understand the practical situation of searching before we can exploit search data for our theoretical agendas—or, more positively, before we can discover new ways of exploiting its potential. Existing

[1] Additionally, there are many hidden algorithmic confounders that undermine both the non-reactive quality and the openness of search data. For example, auto-completion suggests queries to users. While this lowers the barrier to searching, it also means that what we observe may reflect platform-generated prompts rather than the "natural" wording of the user. Auto-completion functions by anticipating what users intend to search for, and if these suggestions are taken up, they shape and alter user behavior in ways that are analytically significant. Taken together, these aspects highlight critical caveats for both qualitative and quantitative analyses of search data.

research on search practices(see Lewandowski 2008; Rieger 2009) is limited and rarely integrated into studies of exploiting search data for studying discrimination(Stephens-Davidowitz 2014), mobilizations(DiGrazia 2017) or infection rates(Ginsberg et al. 2009). As a result, we rely on assumptions about why and how people use search engines—assumptions that may be mistaken, either in general or in the particular case under study.

Social Media data: Social media platforms such as Reddit, X, and Facebook capture our relational lives across both small and large publics. For the first time, we have access to population-level discourse on public concerns, spanning a wide range of issues and settings(Golder and Macy 2014; Lazer and Radford 2017). On the one hand, there are large political publics where climate change might be debated on a politician's Facebook page with hundreds of participants and thousands of spectators. On the other hand, there are small local publics where residents discuss something as mundane as the new opening hours of a neighborhood supermarket. Social media thus provides us with unprecedented data on how ordinary people communicate about issues and with one another. Rather than being limited to the voices of political and cultural elites, we now have direct access to the moral and cultural practices of everyday citizens. As Lazer and Radford (2017) observe, "*Social media have become an essential mode for people at the margins of society to connect with one another and express thoughts and sentiments that otherwise go unsaid.*"

While the growing population of text producers and practices provides an important new data resource for the social sciences, it also poses significant challenges. Different populations speak and write in distinct ways, and texts take on different meanings depending on their context. In many datasets, we have little prior knowledge of which populations and practices produced the texts. Moreover, the sheer size of text corpora makes them impossible to read and analyze manually. Precisely because of this diversity, contextual dependence, and open-endedness, such data must ultimately be understood as highly qualitative.

Scholars have been quick to note the lack of demographic information, the influence of algorithmic confounders, and population drift in social media research. Equally pressing, however, are the interpretative challenges of making sense of what people are actually saying and doing on these platforms. Compared to survey data, the differences are immediately apparent. In a survey, every respondent encounters the same textual stimuli or questions under standardized conditions. On social media, by contrast, no two contributions are alike. Each comment is made in response to a specific post, follows a particular sequence of prior exchanges, and is authored by someone with their own intentions—whether to create an effect, engage in dialogue, amuse friends, or address the original poster. This context is far removed from the controlled environment of surveys, where question wording and order are carefully standardized to enable comparison.

At the same time, social media data offers advantages over other forms of digital trace data, such as search queries. Whereas search data often consists of isolated words or short phrases, social media typically provides full posts or comments together with their surrounding conversational context. This results in a much higher degree of datafication of the communicative situation and produces an open-ended data form well suited to qualitative analysis.

Sensory Data: Mobile phones and other tracking devices capture the rhythms of our everyday movements and interactions in physical space. For the first time, researchers can observe population-level patterns of mobility and co-presence across diverse settings(Alessandretti, Aslak, and Lehmann 2020; Sekara, Stopczynski, and Lehmann 2016; Stopczynski et al. 2014). On the one hand, such data can reveal large-scale flows, such as commuting patterns across a metropolitan area or the spread of people through transport hubs. On the other hand, it can illuminate small-scale gatherings, such as a group of students meeting regularly in a

campus café or friends attending a local concert together. Sensory data thus provides a window onto the spatial and temporal organization of social life—how individuals move, assemble, and overlap in shared environments.

Yet these new opportunities also present major challenges. A single GPS or bluetooth trace might represent a daily commute, a spontaneous encounter, or a meaningful social ritual, but the data alone rarely discloses which. The same location coordinate may capture a solitary walk, a romantic meeting, or a collective protest. As with social media, researchers often lack prior knowledge about the social situations that generated the traces, and the sheer scale of sensory datasets makes close, manual interpretation infeasible. Precisely because of this contextual dependence, ambiguity, and situational embeddedness, sensory data must be understood as highly qualitative—even when it appears as numerical coordinates or proximity signals.

Compared to survey or administrative data, the interpretive challenges are stark. A survey typically situates a respondent's answer within a known context, but a phone sensor reading only tells us that two devices were near each other, not whether their users were interacting, ignoring one another, or merely standing in line at the same shop. This thinness of information makes sensory data simultaneously powerful and opaque: powerful because it provides continuous, large-scale observations of human movement, and opaque because interpretation always requires additional contextual knowledge.

At the same time, sensory data also holds unique advantages over other digital trace sources. Unlike social media or search queries, which rely on self-expression, sensory data often captures behavior without conscious effort on the part of users. This unobtrusiveness can reveal aspects of social life that people may not articulate or even notice themselves, such as habitual routines or tacit forms of coordination. But precisely because these traces are both thin and situated, their meaning cannot be taken for granted—they demand qualitative interpretation to transform numbers and coordinates into accounts of social life.

4. Four qualitative tasks in digital trace analysis

The three data sources we have highlighted—social media, search queries, and sensory traces—illustrate the diversity of digital trace data. Each provides unprecedented access to dimensions of social life that were previously difficult, if not impossible, to observe at scale: discourse in public and semi-public arenas, the inner concerns expressed through private searching, and the embodied routines of mobility and co-presence. At the same time, each comes with distinct challenges of interpretation. Social media data is rich but unruly, full of contextual dependencies and platform-specific conventions. Search data is highly unobtrusive but thin, offering only fragments of intent that resist straightforward interpretation. Sensory data is continuous and large-scale but opaque, demanding auxiliary information to situate what a given trace actually represents.

Taken together, these cases underscore a key point: digital trace data may differ in form and degree of datafication, but in all instances, their meaning depends on the contexts and practices from which they emerge. This makes qualitative research strategies an indispensable part of social science methods in the digital age. To move from raw traces to valid social science knowledge, researchers must employ strategies that help them specify what kind of case they are dealing with, ground their interpretations empirically, generate theories that account for unexpected findings, and validate their quantitative measures with qualitative analysis. In the following, we therefore outline four qualitative tasks—casing, grounding, theorizing, and

validating—that we consider productive for making sense of digital trace data and for ensuring that the social sciences can responsibly and productively reappropriate these new data sources.

4.1. Task 1: Casing

By casing, we mean answering the question "what is this a case of?" through empirical investigation. In qualitative social science, casing is a central analytical move that involves specifying the phenomenon under study by situating it within a broader conceptual or empirical category (Becker 2024; Krause 2024; Ragin 2009; Ragin and Becker 1992; Soss 2021; Tavory and Timmermans 2014). Rather than treating cases as mere illustrations of pre-existing theories, qualitative researchers often begin with open-ended inquiry and allow the meaning of the case to emerge through analysis. This process involves iteratively refining the case boundaries, comparing with other instances, and adjusting theoretical frames to better capture what is distinctive or generalizable about the case. Casing is thus not just about selecting cases, but about constructing them in dialogue with theory and data, enabling researchers to clarify what kind of social process or phenomenon their data actually represent.

The first argument for *casing* is that the digitalization of social life transforms social processes, requiring us to examine the extent to which our existing conceptual frameworks need to be revised(Marres 2017, 2020). A clear example is the concept of public opinion. While public opinion is undoubtedly formed and expressed on social media, it does not align neatly with the dominant conception of public opinion. Social media populations are not representative of national publics, which are typically the object of study in traditional public opinion research. Moreover, the public articulations that occur on social media do not necessarily reflect individual opinions in a stable or isolated sense, but rather situated behavior in specific contexts. Therefore, social scientists must engage in casing to correctly specify the object of study, adapting their concepts to better fit the empirical realities of digitally mediated social life.

The second argument for *casing* is to enable qualified generalization. Digital trace data can be described as *large-particular-N data*—large in terms of the number of people included, but particular in terms of the population and practice to which it refers.Traditional surveys typically target a broadly defined population, such as a national population. When inferences are made to this population, the results are usually considered general—although one could, and many have, argued that such surveys are themselves national or historical case studies. In contrast, the populations captured through digital trace data rarely have this quality of broad generalizability. It is often highly relevant to ask whether the findings apply to other samples, platforms, or topics. Therefore, to make results from digital trace data more broadly relevant, researchers must engage in case-based generalization. One approach is to relate a given case study to other cases within the same class of phenomena and then use qualitative knowledge of those cases to make a reasoned argument about the generality of the findings. Flyvbjerg, among others, has argued that social processes can be generalized from cases when the type of case analysed is specified. Most-likely and least-likely cases, in particular, can be used to extend insights: if a social process proves consequential, or an opinion prevalent, even in a least-likely case, we can infer that it will be so in more likely cases as well (Flyvbjerg 2006).

4.2. Task 2: Grounding

4.2.1. By grounding, we mean that our theoretical concepts and interpretations are continuously informed by and put to the test by the practices and meanings of those we study. One way to answer the question of why ethnographers spend months, even years, doing fieldwork is that they seek to properly ground their

interpretation and analysis of their research site. In qualitative research, grounding is typically achieved through close engagement with empirical material—such as interviews, observations, or textual data—and through iterative analysis of how the actors studied themselves make sense of their world(Glaser and Strauss 2017; Lichterman 2017; Malinowski 1994). Rather than applying fixed categories or a fixed research design researchers adjust their concepts and empirical investigation in light of what they encounter in the field. This ensures that interpretations remain empirically adequate and sensitive to context. In the following, we address three forms of grounding that are particularly important when working with digital trace data: the grounding of the researcher, the grounding of the empirical data, and the grounding of the theoretical concepts.

4.2.2. The first type of grounding is the grounding of the researcher, making the researcher a qualified interpreter of digital traces. We can't assume to know what people mean, what they are doing, and how they see the world. A core principle in qualitative research is that researchers must spend time with their material—reading, observing, listening—so they become competent interpreters of the social setting that they are studying (Lichterman 2017). As Small and Calaraco argue, extensive exposure is a prerequisite for high quality qualitative evidence (Small and Calarco 2022). In the context of digital trace data, this means engaging directly with the content: reading social media threads, exploring user interactions, observing people's movement and co-location patterns, reading search strings. In the context of social media data, such immersion enables researchers to grasp platform-specific conventions, tone, irony, relational dynamics, and other subtle features that might otherwise be misinterpreted. Grounding the researcher in this way is not only about familiarity with the data but also about acquiring the interpretive competence necessary to make contextually informed judgments about meaning and relevance. The extent to which qualitative strategies, such as open and focused reading(and coding), is a useful strategy for grounding the researcher depends on the extent to which the digital data source is open in its data format and the degree of digitalization of the social situations. A researcher can learn a lot from reading social media posts about how people communicate, but less about how people interact through sensory mobile data. In order to really understand what the sensory traces mean in terms of social interaction it is necessary to combine the digital trace data with in-situ observation of the social gatherings (Blok et al. 2017; Stopczynski et al. 2014).

The second type of grounding is grounding of the data—ensuring that researchers have the relevant contextual information necessary to interpret any given observation. Take, for example, sensor data from phones: while we might assume that proximity between two devices indicates interaction between individuals, this assumption may only hold in certain settings and not in others. In such cases, additional data—such as the location—may be required to interpret the meaning of the sensor data accurately. A similar issue arises with social media comments or posts: to what extent can a given utterance be understood without information about the surrounding discourse or the actors involved? This depends on whether the relevant context is made explicit in the text. For example, a short comment like "Sounds cosy. Who is paying?" is difficult to interpret in isolation. The first part could be ironic or sincere, and the second could be a critique of excess or a genuine question. However, if the comment appears under a post stating, "The Prime Minister welcomes 42 former and current female ministers for lunch," it becomes clearer that the comment is a critique of wastefulness. It is not always possible to know in advance what context is needed to infer the meaning of an utterance. Qualitative researchers address this through context tracing—investigating the surrounding circumstances of a specific observation on a case-by-case basis. This is why follow-up questions are so crucial in in-depth interviews and ethnographic fieldwork: they allow researchers to quickly gather clarifying information when an action or statement is initially ambiguous (Lareau 2021; Weiss 1995). Similarly, in digital research, the researcher can attempt to reconstruct or trace the relevant context in each case. In

large-scale data analysis, it becomes important to consider what contextual information is required to make valid inferences about what a digital trace means. For example, when training or validating a machine learning model, what contextual information is needed—both by the computational model and the human annotator—to properly understand a social media message?'

A third type of grounding is the grounding of concepts, ensuring that each concept is specified in a way that is empirically adequate. We borrow the notion of grounding from Grounded Theory, originally proposed as a method for generating both concepts and integrated theories through close empirical investigation. In Glaser and Strauss's classic formulation (2009), Grounded Theory is not about deducing concepts from prior assumptions but about allowing concepts to emerge systematically from data analysis. Deductive approaches, they argue, often result in opportunistic uses of theory that do not fit the data. By contrast, concepts in Grounded Theory are gradually developed and refined "in relation to the data during the course of research" (Glaser and Strauss 2009:6). Conceptual grounding therefore requires an iterative process, where the adequacy of each concept is continuously tested against empirical material. A central strategy here is theoretical sampling: selecting cases with respect to the development of specific concepts or categories. The purpose of such sampling is not empirical generalization but theoretical saturation—refining concepts until further empirical variation no longer alters their content (Small 2009). While Grounded Theory proposed a radical inductivism, where concepts should be derived solely from data analysis, others have argued that concepts are typically taken from prior theories and reworked in new empirical settings (Tavory and Timmermans 2014). Nevertheless, the point remains that concepts must be worked out, tested, and reformulated in dialogue with empirical material.

4.3. Task 3: Theory generation

Going through the exploratory phase of theory generation enables researchers to use digital data's theoretical inadequacy to generate theory. To illustrate, consider an example from our own research. We have collected a large Facebook dataset containing posts from Danish political actors, along with users' likes and comments on these posts. The first question we asked ourselves was: *What could this data be seen as a case of?* One obvious interpretation is political participation. When Facebook users like or comment on a politician's post, they engage in a form of political behavior akin to writing letters to politicians or signing petitions. In this reading, the data provides individual-level, continuous, observational indicators of political participation for all citizens interacting with politicians on Facebook.

However, this interpretation is not without challenges. Liking and commenting on social media are often low-cost, reactive, and loosely organized actions that may lack the weight traditionally associated with political participation. Critics have dismissed such practices as "clicktivism," suggesting that they create an illusion of activism rather than genuine engagement (Lewis, Gray, and Meierhenrich 2014). Another common interpretation is to treat likes and comments as expressions of public opinion (Murphy et al. 2014). From this perspective, reacting to a politician's post resembles answering a survey item, since both are "opinion statements" formulated in response to an issue (Zaller 1992) . Social media trace data might then be seen as a low-cost substitute for traditional survey data. Yet this framing also faces critique: social media publics are not representative of national publics, and serious selection biases exist in both who participates and what gets expressed (Conrad et al. 2019, 2021). Thinking across these two conceptualizations suggests a third possibility: to interpret likes and comments directed at political elites as expressions of *mobilized public opinion*. Rather than simply indicators of individual participation or static opinion, such traces can be seen as evidence of publics that actively mobilize their views through social media. This reconceptualization not only better

captures what is happening on these platforms but also opens new lines of inquiry: Who mobilizes their opinions? By what mechanisms? And with what effects on political elites? Importantly, this example illustrates how the ambiguities and selection problems that complicate standard interpretations of digital trace data can themselves become a source of theoretical innovation, provided that researchers engage in the kind of active concept work and theorizing we argue is central to qualitative approaches.

4.4. Task 4: Validating

Most research drawing on large-scale digital data seeks to quantify some aspect of social life and conduct statistical analysis. Since digital data sources are typically qualitative in nature, this requires qualitative validation before we can trust and apply the resulting quantitative measures. The difficulty of validating quantitative measures with qualitative methods varies depending on the openness of the data situation and the degree of its digitalization. In this section, we first consider how to validate social media text classification, which represents an open and highly digitalized case, and then turn to sensory data from mobile phones, which represents the opposite. We limit ourselves to discuss the validation of quantitative measures and do not address the validation of results in a broader sense.

Researchers have used large-scale text data to study political issue dynamics (Barberá et al. 2019; Ennser-Jedenastik et al. 2022), global individual level sentiment patterns(Golder and Macy 2011) and political logics within social movement (Nelson 2021) and much more. A dominant approach has been the use of unsupervised methods, especially topic models, to locate and measure themes in text. While it has long been recognized that measures derived from these models must be validated, this has primarily been done through face validity—did the categories make sense to the researcher—and, in some cases, through indirect validation by correlating them with other variables. Yet, such strategies do not safeguard against systematic measurement error (Carlsen and Ralund 2022). With the rise of supervised learning and large language models for classification, there has been a shift toward direct validation of text classifications using test sets, where a sample of the model's predictions is compared against a human-annotated sample of text functioning as the ground truth. This allows for direct evaluation of measurement error. However, this move to direct validation makes the role of human annotators critical for assessing model performance. As noted above, we cannot assume that annotators will reliably interpret textual messages — especially in the often ambiguous and context-dependent domain of social media. For annotation to provide trustworthy ground truth, it must be carried out by qualified interpreters who are grounded in the field site and relevant theoretical frameworks.

While the cost of grounding annotators and labeling texts is significant, it is nonetheless relatively straightforward to sample digital text and use it to evaluate quantitative measures. This is not the case for digital data sources that are closed in format and have a low degree of digitalization, such as sensory data from mobile phones. If we use proximity data to measure social relations or active interaction, we must verify whether co-location of phones actually corresponds to people interacting. The challenge is that we cannot draw a random sample of phone-to-phone encounters and observe them in situ to assess whether a detected link truly represents an interaction. One approach to evaluation is to focus on a specific place and time, as in Blok et al. 2017, where measures derived from mobile sensory data were cross-validated against an ethnographer's field notes from the same event—an on-campus party. Yet, the extent to which validation from such a particular event can be generalized to other contexts remains highly questionable.

Here, case selection logic can help researchers choose contexts (in this case, particular locations at particular times) where situ observations can be carried out to distinguish between bodies that are interacting and those

that are merely in proximity. If such validation sites are carefully chosen, estimates of measurement error can be generalized to the broader dataset with greater confidence. Generalizations must, however, remain approximate. For example, if 25 percent of co-location events in a most-likely setting are identified as false positives, one can reasonably infer that error rates in less problematic contexts are lower, thereby providing a bounded estimate of overall measurement accuracy.

Having outlined four key qualitative tasks—casing, theorizing, grounding, and validating—we now turn to two extended case studies that illustrate these dynamics in practice. Both draw on large-scale digital trace data but highlight different challenges of interpretation and validation.

5. Applying qualitative research strategies to digital trace data in different types of data situations.

In this section, we illustrate how the four research strategies can be applied when working with digital trace data by drawing on two of our own research projects: one representing a case with a low degree of datafication, and the other a case with a high degree of datafication.

5.1 Understanding work sociality using sensory data and qualitative interviews

In an ongoing case study of hybrid and flexible workplaces, we examine the use of free seating to study relational preferences, based on the assumption that choosing whom to sit next to reflects a social preference for that person. Put more simply, we treat co-seating as a tie in a social network analysis. We will attempt to illustrate how grounding and validating can be done using traditional qualitative methods in data situations with a low degree of digitalization.

The digital trace data is collected from *DeskBooker*, an app used in a large Danish finance company where employees alternated between remote and office work. At the company's headquarters, 754 employees shared 420 desks, and each day they worked on site, they were required to use *DeskBooker* to reserve a workstation. Our dataset comprises 33,000 bookings across nine open-plan offices over a six-month period. Each booking contains the name of the employee, the time of booking, the booked date, and the exact coordinates of the chosen desk. From this data, we are able to construct a network of ties between employees (see Figure 1).

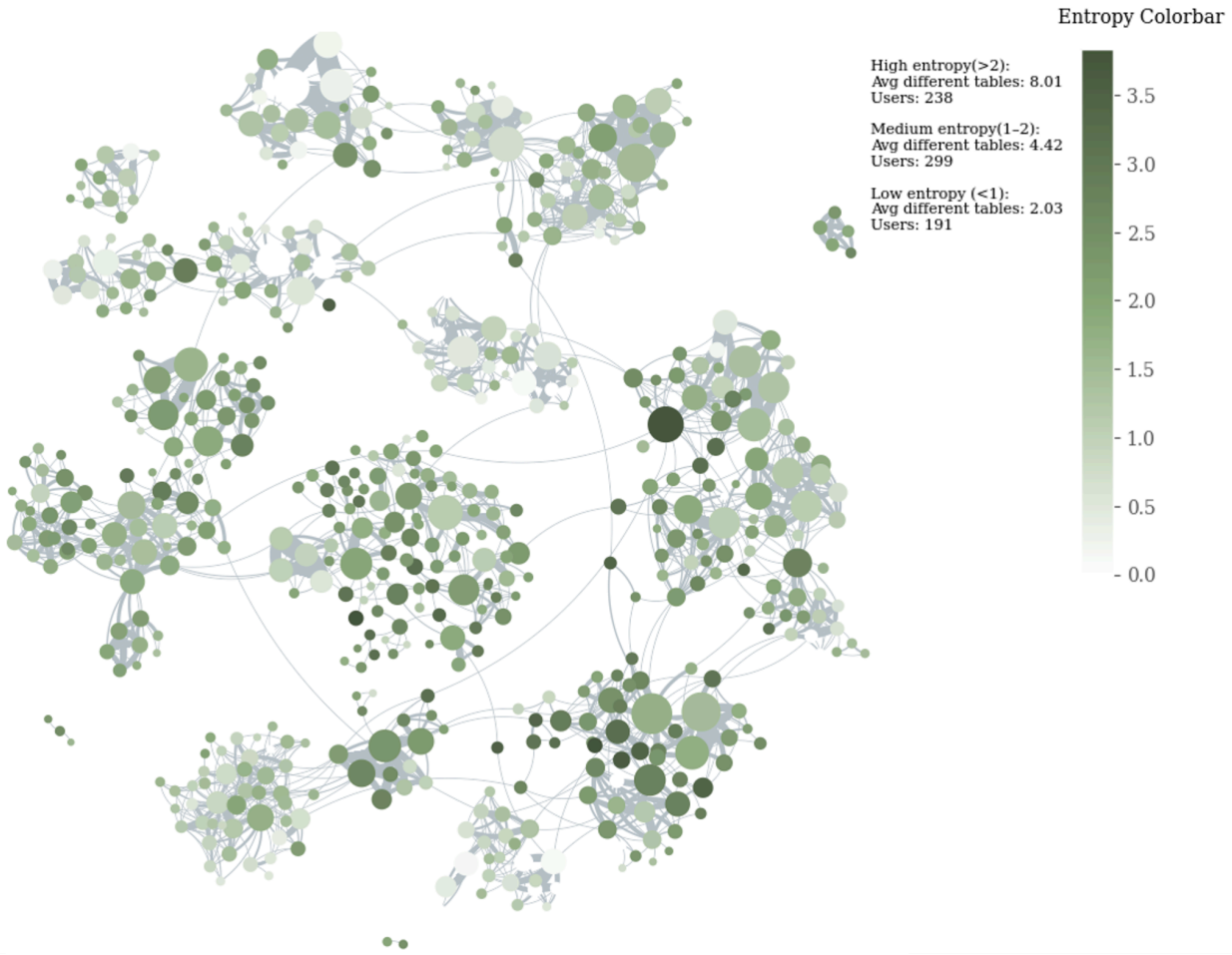

Figure 1: Network of employee relations based on co-presence.
Each node represents an employee. Two employees are connected if they have booked neighboring desks at least three times over a six-month period. Thicker edges indicate more frequent co-bookings, and node size indicates booking frequency, i.e., how often an employee goes to the office. Node color intensity represents desk entropy, meaning variation in desk choice: darker colors indicate that an employee books many different desks, whereas lighter colors indicate that they consistently book only a few. The network is highly clustered, showing that employees most often sit with the same few colleagues. Employees with high desk variation tend to sit with people across different parts of the company. The average number of connections per employee is 5.2.

Such a dataset resembles other forms of sensory data in that it allows observation of employees' patterns of mobility and co-presence whenever they attend the office. As previously noted, sensory data typically has a low degree of datafication, meaning that we know relatively little about the social situations underlying the desk bookings. This makes it difficult to draw meaningful conclusions from booking patterns based solely on quantitative analysis. Do these traces genuinely capture relational preferences between individuals, or are there other reasons why two people end up sitting next to each other?

To address this uncertainty, one author conducted interviews and participant observations, generating insights essential for interpreting the dataset. Hands-on experience with *DeskBooker* during participant observations clarified several of the affordances and restrictions programmed into the app, thereby shaping the technical side of the data situation. When employees opened *DeskBooker*, they encountered a detailed map of the workplace showing available desks for specific days, while occupied desks displayed the names of those who had already booked them. This feature proved critical for understanding the social side of the data

situation—namely, the practices through which bookings were made. Although employees were formally allowed to sit anywhere, the app enabled them to choose precisely where, and with whom, they wanted to sit in the office. Interviews revealed that employees often used the app strategically to book desks near close colleagues, or alternatively chose to work from home when this was not possible.

> *When you come in, of course you want to sit with the people you know, and in our group, we use that feature a lot. The fact that you can see where your closest colleagues are sitting and then book a spot nearby. (…). That's one of the advantages of going [to the office]: being physically together with your colleagues when you are there, and being able to choose that.*
>
> - Employee talking about using DeskBooker

This evidence supported our initial assumption that desk booking was driven by relational preferences, yet other interviews revealed an additional factor shaping booking behavior. Many interviewees reported that desks near hallways, meeting rooms, or in the middle of a room were less attractive than those along walls or near windows, due to distractions and a sense of being surveilled. The interviews therefore served both to confirm our interpretation of desk booking as an indicator of relational preference and to highlight an alternative explanation: spatial preference. In the final analysis of this network, we therefore accounted for "avoidance zones" within the workspace—areas where employees rarely chose to sit or interact. In this way, the qualitative insights clarified the social situation underlying the *DeskBooker* data and enabled a more meaningful interpretation of the quantitative results.

Overall, the relational and spatial preferences underlying employees' booking behavior emerged as an insight that quickly reached saturation, as nearly all interviewees described the same thought process when explaining their choices. This suggests that, in this case, only a relatively small number of interviews was needed to provide sufficient grounding of the digital traces. It is worth noting, however, that this efficiency is partly due to the highly bounded nature of the data situation. Unlike other forms of sensory data, which may track mobility across diverse settings within a bounded population, or combine unbounded contexts with unbounded populations, desk-booking data is confined to a single workplace. The more bounded the geography and population, the more likely it is that the data situation is shared across the sample—meaning that a smaller number of interviewees can provide adequate support for within-case generalization.

5.2 Understanding non-contentious mobilization using social media data and mixed methods text analysis

In the following section we will exemplify how qualitative inquiry helped with three tasks 1) generating theory in light of a suppressing null finding 2) and in turn how casing can be used to critique the theoretical proposition, 3) how qualitative analysis helps validate computational text measures. We will take the Danish refugee solidarity movement mobilization on Facebook in 2014-2016 as our case.

The surprising empirical finding was the Danish refugee solidarity movement's persistent lack of contention. Standard theoretical approaches would predict that mobilization around refugee rights during a period of electoral threats and restrictive immigration policies should have turned contentious (Benford and Snow 2000; Diani 1992; Gamson 1992). One obvious explanation for this finding is that the mobilization focused solely on crisis relief and was unconcerned with broader social change. Yet extensive qualitative analysis of both the groups' organizational self-descriptions and their everyday interactional practices revealed a powerful non-contentious frame, crystallized in the movement's largest faction, *The Friendly People*. In this frame, social

change was envisioned not through protest or blame attribution but through hospitality, everyday sociability, and positivity. Board games, bicycle rides, and coffee gatherings with refugees were framed as "bridge-building" actions that created joy and mutual respect, while negativity and political talk were explicitly discouraged. The problem was defined not as state injustice but as a lack of friendliness; the solution lay in cultivating hospitable interaction; and the motivation was the promise of reciprocal happiness. From this emerged the theoretical proposition that non-contentious collective action frames can compete with, and even undermine, contentious frames in cycles of mobilization.

Our focus on the importance of non-contentious frames within a movement oriented toward social change relies critically on how the phenomenon is *cased*, as alluded to above. When we call the mobilization a *social movement mobilization* rather than simply a *crisis relief effort*, the finding of non-contention becomes puzzling; without that framing, it does not. A possible critique of this case study—and of much of the first author's work on the mobilization—is that the casing is simply wrong, which would render the framing of the studies, their theoretical expectations, and subsequent theory development ill-conceived. In this case, we defended the casing by drawing on the discourses of movement elites, organizational documents, and the everyday practices of rank-and-file members, all of which pointed to the presence of a social change–seeking collective action frame. Still, our large-particular-N study, like many digital trace research projects, should be tested and criticized in terms of its casing, since it is this decision that gives many findings their significance and relevance—much more so than statistical tests, measurement strategies, or model performance. To critique (and defend) casing, however, requires either conducting qualitative research or having access to qualitative material from which to draw conclusions. Hence the importance of including a background section that details the case in sufficient depth to allow others to assess the adequacy of the casing.

To be certain that the movement was genuinely non-contentious, we went beyond examining organizational and elite statements and sought to measure contention within the Facebook groups of rank-and-file members. We were interested both in the overall share of contentious posts and comments and in how this changed over time in response to an increasingly hostile political environment. To this end, we classified all 640,000 posts and comments collected from 119 Facebook groups as either contentious or not. This computational measure relied heavily on qualitative analysis in several ways. First, our qualitative analysis led to a reconceptualization. We began with Boltanski and Thévenot's concept of critique, which requires that a statement be qualified in some way—that is, that an actor formulates a moral standard to justify their disapproval[2]. Yet after reading thousands of posts, we found very few instances of such explicit critique. To adequately capture discontent with the political system or with public attitudes toward refugees, we therefore broadened our definition of contention to include both explicit and implicit critiques. Second, in order to train and test our supervised machine learning model, we needed to label posts and comments whose meaning was often hard to interpret. For this reason, the two annotators chosen were those with the greatest qualitative exposure to the movement, as their immersion in its discourse enabled them to classify the texts more accurately. Hence, the model was not trained or tested against just *any* human annotator, but against qualified annotators who had themselves gone through a long process of learning, testing, and revisiting their understanding of the movement's discourse. Although more costly, training and validating a text classifier with qualified human annotators has the major advantage of reducing the likelihood that misclassifications stem

[2] While it is true that starting with a theory or multiple theories runs counter to many versions of grounded theory, we do not consider strict adherence to grounded theory to be particularly productive when working with digital trace data. Instead, abductive approaches, as formulated by Timmermans and Tavory (2014), provide a more effective way of engaging theoretically with digital trace data (see also Carlsen, 2026). What is crucially retained from grounded theory—and shared with abductive analysis—is the insistence on theorizing through sustained empirical analysis.

from the annotators. This, in turn, allows researchers to focus on improving the model rather than questioning the validity of the ground truth.

More broadly, this case illustrates the analytical value of qualitative inquiry for interpreting substantively meaningful null findings. In recent discussions of transparency and replicability, null results have increasingly been recognized as important scientific outcomes rather than as mere failures to detect effects. Yet digital trace research often lacks tools for making sense of such findings when they contradict theoretical expectations. We argue that casing and theory generation are central to addressing this gap. Through careful casing, researchers can establish why a null finding is theoretically puzzling in the first place, while theory generation allows them to reformulate expectations in light of empirical evidence. In this way, qualitative inquiry renders null findings theoretically productive in a transparent and empirically rigorous fashion, by revealing the social meanings, interpretive frames, and organizational logics that underlie the absence of expected patterns.

Taken together, these two examples demonstrate the indispensable role of qualitative inquiry in making sense of digital trace data. Whether working with highly digitalized sources such as social media text or with minimally digitalized sensory data, the same four qualitative strategies—casing, theorizing, grounding, and validating—proved essential for transforming raw traces into credible social science findings. In one case, qualitative immersion enabled the articulation of a new theoretical proposition in response to a surprising absence of contention; in the other, it provided the only means of assessing whether seating measures genuinely reflected relational preferences. These cases thus underscore that qualitative engagement is not an optional supplement to computational analysis but a necessary condition for generating valid measures and meaningful interpretations. With this in mind, we now turn to the conclusion, where we draw together these lessons and consider their broader implications for qualitative approaches to digital trace data.

6. Conclusion - Everything all at once or when we need to do what tasks

In this paper, we have argued that digital trace data, despite its scale and technological mediation, is in many respects qualitative data. What makes such traces meaningful are the specific situations in which they are generated, and it is only by attending to these data situations that researchers can responsibly reappropriate them for social science. Our proposal has been to articulate four qualitative tasks—casing, grounding, theorizing, and validating—that are indispensable for turning raw traces into robust knowledge. Each task addresses a distinct danger: mis-specifying the object of study, misinterpreting observations, missing theoretical opportunities, or misplacing trust in measures. Taken together, they form a repertoire of strategies that allow researchers to engage critically and productively with large-scale digital infrastructures.

Yet not all traces demand the same kinds of work. Different data situations, defined by the openness of their formats and the degree of datafication of the underlying social processes, call for different emphases in qualitative strategy. In the case of social media, where data is relatively rich and open, the most pressing tasks are casing and grounding: we must specify what a stream of comments actually represents and immerse ourselves in the communicative conventions that shape them. With search data, by contrast, the thinness and ambiguity of the traces make grounding and validation paramount. Here, additional qualitative data—such as interviews, diaries, or contextual studies of search practices—are often indispensable if we are to know what a

query really means. Sensory data, finally, confront us with the opacity of behavioral traces such as GPS pings, Bluetooth proximities or other locational traces. These require extensive contextual grounding and qualitative validation, often through observation or interviews before they can be credibly mobilized as indicators of social interaction.

The point, then, is not that researchers must always do everything at once, but that they must always do the right qualitative thing next. When the data situation is open and richly contextual, the traces themselves may be sufficient for casing and theorizing. When the situation is thin or closed, additional qualitative data is needed to supply context, test assumptions, and stabilize meaning. When surprising results emerge, theorizing becomes indispensable; and whenever qualitative traces are translated into quantitative measures, validation is essential.In this sense, qualitative approaches are not optional supplements to the digital data revolution but constitutive of it. The promise of digital trace data lies not simply in its unprecedented scale but in its continuity with the long-standing qualitative concern for situated practice and meaning. To realize that promise, researchers must learn to move flexibly between tasks: to read traces closely, to ground them in context, to theorize from their surprises, and to validate their translations. Only then can digital traces serve as a cumulative resource for the social sciences—one that expands rather than contracts our understanding of social life.

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